An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

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Project Proposal Report

Wickremathilaka J A– IT22152114

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Wickremathilaka J A	IT22152114	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Thilini Jayalath	20/03/2026	

ABSTRACT

Sri Lanka’s transition toward a USD 15 billion digital economy is hindered by a critical disconnect between tertiary academic outputs and labor market requirements. While the nation maintains a high digital literacy rate, functional computer literacy the ability to perform professional technical tasks dropped significantly to 35.9% in 2024, creating a workforce that lacks the specific competencies demanded by the modern ICT-BPM sector. Current career guidance platforms are often "black-boxes," offering static job matching without explaining the underlying reasoning or identifying the specific "distance" a student must bridge to become market-ready.

This research proposes an **Explainable AI (XAI)** framework for an **Adaptive Career Pathway and Skills-Gap Recommender System**. The solution introduces a **Dynamic Skills Ontology (Knowledge Graph)** to identify skill adjacencies and infer hidden talents. Utilizing a **Stacking Ensemble** of Random Forests, SVM, and Deep Neural Networks, the system calculates a personalized **Skill Gap Index (SGI)** and a structured **"Readiness Timeline"**. Decision transparency is ensured through **counterfactual reasoning**, providing "what-if" explanations to guide student choices.

Technological implementation leverages a **Python-based analytical microservice**, **Node.js orchestration**, and a **React trilingual interface**. Expected outcomes include a **25% reduction in career path anxiety** and an **85% actionability score** for skill-bridge suggestions. Socially, the system mitigates regional and gender biases using **Statistical Parity Difference (SPD)** auditing. Commercially, the platform acts as a **data-driven curriculum alignment tool** for universities, allowing them to identify widespread student skill gaps and update their teaching modules to match real-time industry demands as a **scalable B2B service**.

Keywords: Explainable AI (XAI), Skill-Gap Index (SGI), Career Intelligence, Dynamic Skills Ontology, Multi-Objective Optimization.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
AMOPSO	Adaptive Multi-Objective Particle Swarm Optimization
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
B2B	Business-to-Business
DI	Disparate Impact
DNN	Deep Neural Network
ICT-BPM	Information and Communication Technology - Business Process Management
IEEE	Institute of Electrical and Electronics Engineers
KNN	K-Nearest Neighbors
LSA	Latent Semantic Analysis

Abbreviation	Description
ML	Machine Learning
NLP	Natural Language Processing
SaaS	Software as a Service
SGI	Skill-Gap Index
SLASSCOM	Sri Lanka Association for Software and Service Companies
SPD	Statistical Parity Difference
SVM	Support Vector Machine
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Background and Context

The rapid evolution of the global digital economy, driven by Artificial Intelligence (AI) and automation, has created a significant disparity between academic qualifications and professional competency. In the context of Sri Lanka, the national goal to reach a USD 15 billion digital economy by 2030 requires a workforce of 300,000 highly skilled IT professionals; however, there is a systemic failure in the current education-to-employment pipeline.

The Real-World Problem Undergraduates and job seekers in Sri Lanka frequently struggle with "Career Path Anxiety" due to a lack of data-driven intelligence regarding their market readiness. While students may complete degrees, they often lack the specific, applied skills such as Cloud Computing, DevOps, or specialized AI workflows demanded by the modern ICT-BPM sector. This results in graduates being "paper-qualified" but functionally underprepared for high-value roles.

Who is Affected and Why it is Important This problem primarily affects final-year undergraduates, career changers, and educational institutions. For the student, a lack of guidance leads to underemployment or migration to misaligned career tracks. For the nation, it leads to a "brain drain" and a failure to capitalize on the global digital boom, threatening Sri Lanka's position as a safe and competitive hub for technical talent.

Statistical Evidence (Local/Global)

- **The Literacy Paradox:** While Sri Lanka reports high general digital engagement, functional **computer literacy dropped to 35.9% in 2024**, highlighting a widening gap in professional technical skills.
- **Employability Gap:** Global studies indicate that 85% of the jobs that will exist in 2030 haven't been invented yet requiring a shift from static learning to dynamic skill-gap bridging.
- **Recruitment Friction:** Local employers report that over 70% of candidate resumes do not match the actual technical task requirements of the role, leading to high operational costs in screening.

Analysis of Existing Solutions and Their Limitations

Current solutions, such as standard job portals (LinkedIn, Glassdoor) or university career centers, face several critical limitations:az

- **Static Information:** These platforms provide yearly reports or general advisories that lack the real-time, predictive capability needed to guide individual career tracks.
- **"Black Box" Matching:** Existing algorithms match keywords but fail to explain the logic behind the match or provide a quantified "distance" the student must bridge.

- **Lack of Localized Context:** Global platforms do not account for the specific skill adjacencies or the trilingual (Singlish) nuances of the Sri Lankan workforce.

The Proposed Solution: Adaptive Career Intelligence To address these gaps, this component introduces an **Explainable AI (XAI) Framework for Adaptive Career Intelligence**. The system consists of:

1. **Dynamic Skills Ontology:** A knowledge graph that infers hidden talent by mapping skill relationships rather than just keywords.
2. **Skill-Gap Index (SGI) Engine:** A mathematical model that quantifies the "semantic distance" between a student and a target role.
3. **Explainable Recommender (XAI):** A module that provides human-readable "what-if" insights to help students understand exactly how to improve their readiness.

Relevant SDG and Research Cluster Alignment This research aligns with **SDG 4: Quality Education** by promoting technical and vocational skills for employment, and **SDG 8: Decent Work and Economic Growth** by bridging the youth employment gap. It is situated within the **AI and Machine Learning** research cluster, focusing on predictive modeling and human-centric intelligent systems.

1.2 Literature Review

The following literature review explores scholarly advancements in recommendation systems, explainable AI (XAI), and skill-gap analysis to provide a comparative context for the proposed "Adaptive Career Intelligence" framework.

1.2.1 Person-Job Recommendation Systems (PJRS) and "Black Box" Challenges

Person-job recommendation systems (PJRS) are designed to match candidates with suitable roles by analyzing skills, experiences, and interests [1]. However, recent studies highlight the "black box" nature of complex algorithms such as CNNs and Transformers which obscures how specific features influence a recommendation [2]. This invisibility in feature extraction often hinders user trust and makes bias detection difficult [1, 2]. This study addresses these challenges by moving beyond simple keywords matching to an interpretable decision-logic framework.

1.2.2 Knowledge Graphs and Dynamic Skills Ontologies

Knowledge Graphs (KG) have emerged as a powerful tool to address data sparsity by providing auxiliary information through relational connections [3]. In the recruitment domain, recent heterogeneous KG architectures have demonstrated superiority in generating career-motivated course recommendations and textual explanations compared to traditional hierarchical lists [4]. Recent research specifically applies KG to structurally represent subject knowledge and map "predecessor-successor" relationships between skills, proving effective for learning path optimization [5].

1.2.3 Stacking Ensemble Techniques in Predictive Analytics

To enhance predictive accuracy, contemporary research emphasizes the shift from standalone classifiers to ensemble architecture [6]. Stacking ensemble models, which combine multiple "base learners" (such as Random Forest, SVM, and XGBoost) into a meta-model, have demonstrated significantly higher performance often exceeding 91% accuracy in complex professional prediction tasks compared to individual models [6, 7]. Some novel stacking models have even achieved accuracy as high as 98.8% in employee-related forecasting when integrated with Bayesian optimization [8].

1.2.4 Explainable AI (XAI) and Decision Transparency

The role of XAI in recruitment is critical for enhancing transparency and traceability [9]. Recent systematic reviews identify a shift toward using techniques like SHAP and causal reasoning to improve semantic understanding in resume parsing [9]. Studies indicate that providing explanations increases job applicants' perceptions of process fairness and interpersonal treatment, regardless of whether the decision was made by an AI or a human [10].

1.2.5 Algorithmic Fairness and Bias Mitigation

AI-driven recommendation engines can unintentionally amplify social biases embedded in training data [11]. To maintain objective outcomes, current research proposes various fairness metrics, such as **Statistical Parity Difference (SPD)** and **Disparate Impact (DI)** [11]. Effective mitigation strategies include "in-processing" approaches that incorporate fairness constraints directly into the training algorithm to optimize for both accuracy and equitable distribution across demographic groups [11, 12].

Table 1: Comparison of Existing Research and Systems

Reference / System	**Technique Used**	**Focus Area**	**Key Limitation**
Linardatos et al. [2]	CNN / Transformers	ML Interpretability	Focuses on general model logic, not specific career-gap actionability.
Zhu et al. [3]	Knowledge Graphs	Recommender Systems	General mapping that lacks a quantified "Skill-Gap Index" (SGI).
Kumar et al. [6]	Stacking Ensemble	Predictive Accuracy	Achieves high accuracy but operates as a "black box" without user-facing explanations.

Reference / System	Technique Used	Focus Area	Key Limitation
Kim et al. [10]	XAI Vignettes	Applicant Perception	Focuses on how users *feel* about AI rather than providing a roadmap for skill improvement.
Proposed System	**Stacking + KG + XAI**	**Adaptive Guidance**	**Bridges the gap by providing high-accuracy predictions coupled with actionable "what-if" growth plans.**

1.3 Research Gap

The transition from tertiary education to the professional landscape in Sri Lanka is characterized by a significant "Intelligence Void." Despite the availability of career portals and curriculum-based learning, a critical gap remains in the ability to translate individual potential into market-ready competency.

What is Missing in Current Research

The primary missing element in existing literature is a **Unified Adaptive Framework** that combines high-precision predictive modeling with **Actionable Explainability**. While current research excels in accuracy (using ensemble methods) [6, 7] or transparency (using XAI) [9, 10], there is no existing system that integrates these two into a single "Skill-Gap" roadmap. Specifically, the ability to provide **Counterfactual Career Guidance** telling a student exactly which technical lever to pull to change an outcome remains absent in the Sri Lankan ICT context.

Insufficiency of Current Approaches Current approaches are insufficient for the following reasons:

- **Static Mapping vs. Dynamic Growth:** Existing job portals use keyword-matching (TF-IDF or basic NLP) which fails to recognize "Skill Adjacencies" (e.g., a student strong in Java has a natural path to Android development).
- **The "So What?" Problem:** Current AI matchers might give a student a 60% match score for a "Data Scientist" role but fail to explain *why* the score is 60% and *how* to reach 90% through specific micro-credentials.
- **Neglect of Localized Declines:** Traditional models do not account for the specific 2024 decline in functional computer literacy in Sri Lanka, meaning they recommend roles

based on global standards that may not align with the immediate local "Skill-Gap" reality.

Justification of Novelty and Creativity

This research introduces **"Adaptive Career Intelligence,"** a novel approach that moves beyond mere matching to **Prescriptive Career Engineering**. The creativity of this solution lies in two specific innovations:

1.1 **The Skill-Gap Index (SGI):** A unique mathematical metric that quantifies the "semantic distance" between a user's current profile and an industry-standard role profile using Dynamic Skills Ontology.

2.1 **Explainable Growth Roadmaps:** Unlike "Black Box" recommenders, this system uses **Counterfactual Reasoning** to generate "What-If" scenarios. For example: *"If you complete a certification in AWS, your eligibility for a Cloud Engineer role increases by 35%."*

Evidence-Based Argument

The necessity for this novelty is evidenced by the **35.9% functional computer literacy rate** reported in Sri Lanka in 2024. This statistic proves that "General Degrees" are no longer enough; students need a surgical, AI-driven tool to identify exactly where their technical deficiencies lie. Furthermore, the operational cost for Sri Lankan recruiters to filter through thousands of misaligned CVs [1, 11] confirms a massive commercial demand for a system that "pre-validates" and "pre-guides" talent before they even apply for a role.

2. OBJECTIVE

2.1 Main Objective

To design and develop an **Explainable AI (XAI)-driven Adaptive Career Pathway and Skills-Gap Recommender System** as the core component of my **Final Year Research Project**, utilizing a stacking ensemble of machine learning models and a dynamic skills ontology to identify professional competency gaps and provide actionable, prescriptive guidance for **students, undergraduates, job seekers, and career-transitioning professionals** in Sri Lanka.

2.2 Specific Objectives

To achieve the main objective, the following specific, measurable goals have been established for completion within the project duration:

- **Objective 1: Multi-Source Data Acquisition and Preprocessing** To collect and standardize a heterogeneous dataset consisting of 10,000+ job descriptions and 500+ diverse user profiles (including students and career-transitioners), achieving a data consistency rate of over **95%**.
- **Objective 2: Construction of a Dynamic Skills Ontology (Knowledge Graph)** To develop a domain-specific Knowledge Graph that maps at least **1,000 unique technical and transferable skills** to identify "hidden" talent adjacencies and cross-industry career transitions.
- **Objective 3: Development of a High-Precision Stacking Ensemble Model** To build an ensemble model integrating Random Forest, SVM, and DNN that achieves a predictive accuracy of **91–93%** in identifying optimal career paths for users at various professional stages.
- **Objective 4: Implementation of the Skill-Gap Index (SGI) and Prescriptive Logic** To engineer a mathematical framework (SGI) that quantifies the "semantic distance" between current and target skillsets and prescribes a prioritized list of **3–5 specific bridge-learning resources**.
- **Objective 5: Integration of Explainable AI (XAI) for Decision Transparency** To incorporate counterfactual reasoning modules providing **"What-If" explanations**, ensuring that **80% of test users** can clearly understand the logic behind their recommended career shifts.
- **Objective 6: System Evaluation and Bias Auditing** To validate the system using an **Actionability Score** and perform fairness auditing using the **Statistical Parity Difference (SPD)** metric to ensure recommendations are equitable across different age groups and backgrounds.

3. METHODOLOGY

3.1 Research Approach

This research adopts the **Design Science Research (DSR)** approach. This methodology is ideal for IT research as it focuses on the creation and evaluation of "artifacts" (models, algorithms, and systems) to solve identified real-world problems. The research will iterate through the following DSR cycles:

1. **Relevance Cycle:** Identifying the specific career-path anxieties and skill gaps within the Sri Lankan context.
2. **Design Cycle:** Developing the Knowledge Graph, the Stacking Ensemble model, and the XAI modules.
3. **Internal Evaluation Cycle:** Testing the system against performance benchmarks and user feedback.

3.2 Technical Methods and Algorithms

The system is built on a "Intelligence Pipeline" composed of six primary technical phases, moving from raw data to actionable insights:

1. Data Preprocessing and Feature Engineering

Before the AI can learn, the data must be transformed. I will implement:

- **TF-IDF and Word2Vec:** To convert text-based resumes and job descriptions into numerical vectors that the machine can understand.
- **NER (Named Entity Recognition):** Using **spaCy**, the system will automatically extract "entities" such as Programming Languages, Frameworks, and Years of Experience from unstructured text.

2. Dynamic Skills Ontology (Knowledge Graph)

Using **Neo4j**, I will construct a graph where nodes represent skills, roles, and certifications.

- **Relationship Mapping:** Edges represent "semantic distance" (e.g., Python is closely related to Data Science). This allows the system to identify "Transferable Skills" even when a student hasn't held a specific job title before.

3. Stacking Ensemble Learning (The Prediction Layer)

To achieve the target accuracy of 91-93%, a two-tier model architecture will be used:

- **Level 0 (Base Learners):** I will train **Random Forest** (for non-linear relationships), **SVM** (for high-dimensional data separation), and **XGBoost** (for gradient boosting efficiency).
- **Level 1 (Meta-Learner):** A **Neural Network** or Logistic Regression model will then combine these outputs to determine the "Optimal Career Path."

4. Skill-Gap Index (SGI) Engine

This is the core mathematical innovation. I will use **Cosine Similarity** to measure the distance between the User Vector (U) and the Industry Standard Vector (I).

- The formula used is: $SGI = 1 - \frac{U \cdot I}{|U||I|}$
- A higher SGI indicates a larger gap, triggering the prescriptive engine.

5. Explainable AI (XAI) - Counterfactual Reasoning

To move away from "Black Box" models, I will implement **Counterfactual Explanations**.

- **Logic:** The system will search for the smallest change to a student's profile (e.g., adding "Docker") that would change AI's recommendation from "Junior Developer" to "DevOps Engineer." This provides the "What-if" roadmap for the user.

6. Prescriptive Optimization (AMOPSO)

Finally, to handle multiple goals (e.g., high salary vs. student interest), I will implement the **Adaptive Multi-Objective Particle Swarm Optimization (AMOPSO)** algorithm. This ensures the recommended learning path is not just technically correct, but also achievable and rewarding for the specific student.

3.3 Tools and Platforms with Justification

Table 2: Tools and Platforms with Justification

Category	Tool / Platform	Justification
Language	**Python**	The industry standard for AI/ML with extensive libraries like Scikit-learn and TensorFlow.
Backend	**Node.js / Express**	Facilitates high-speed, asynchronous communication between the Python microservice and the frontend.
Database	**Neo4j & MongoDB**	Neo4j handles the complex skill-graph relationships, while MongoDB stores unstructured user profile data.

Category	Tool / Platform	Justification
AI Modeling	**Scikit-Learn / SHAP**	Scikit-learn for ensemble training and SHAP (SHapley Additive exPlanations) for local model interpretability.
Frontend	**React.js**	Provides a dynamic, responsive UI for displaying complex data visualizations like heatmaps and roadmaps.

3.4 Data Collection and Ethical Considerations

3.4.1 Data Acquisition Strategy

To ensure the Skill-Gap Index (SGI) is grounded in the current Sri Lankan economic landscape, the following multi-tier data collection strategy will be implemented.

- **Industry Reference Benchmarks (The "Ideal" Target):** The system will ingest the **2026 SLASSCOM Graduate Personas** and **ICTA Professional Standards**. These documents define the "Target Vectors (I)" for various career paths, representing a 100% market-ready profile.
- **Dynamic Skill Trend Analysis:** Custom web scrapers using **BeautifulSoup and Selenium** will extract data from leading local portals (**Rooster.jobs**, **TopJobs.lk**) and **LinkedIn**. This data is used to identify **emerging technical trends** and update the Skill-Gap Index (SGI) logic so that the career path advice remains relevant to the current year.
- **National Literacy Context:** Data from the **Department of Census and Statistics (DCS)** specifically the *2025/2026 Computer Literacy Survey* will be used to calibrate the recommendation engine based on localized skill availability.

3.4.2 Ethical Considerations

- **Data Privacy & Compliance:** All data collection and storage will strictly adhere to the **Sri Lankan Personal Data Protection Act (PDPA) No. 9 of 2022**.
- **Anonymization:** A rigorous de-identification process will be applied to all user profiles, removing Names, NIC numbers, and Contact Details before the data enters the **MongoDB** environment or the training pipeline.
- **Informed Consent:** For the collection of user profiles, a digital consent form will be mandatory, explicitly stating that the data is used solely for research and career path prediction.
- **Algorithmic Fairness:** To prevent regional or gender bias, the system will be audited using the **Statistical Parity Difference (SPD)** metric to ensure that a student from a rural province receives the same quality of guidance as a student from an urban center [11, 12].

3.5 Validation Strategy and Performance Metrics

The system will be validated using a dual-layered approach:

1. Technical Validation (Offline):

- **Accuracy / Precision / Recall / F1-Score:** Standard metrics to evaluate the Stacking Ensemble's ability to correctly classify career paths.
- **Mean Absolute Error (MAE):** To measure the precision of the Skill-Gap Index (SGI) calculations.

2. User Validation (Online):

- **Actionability Score:** A 5-point Likert scale survey for users to rate how "clear and useful" prescriptive bridge-learning suggestions were.
- **System Usability Scale (SUS):** To evaluate the overall user experience of the career roadmap interface.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

The proposed component is architected as an autonomous **Intelligence Engine** designed to transform abstract professional profiles into quantifiable career roadmaps. Unlike traditional search-based systems, this architecture is grounded in a **Machine Learning (ML) Lifecycle** that bridges the gap between raw labor market data and individual career guidance.

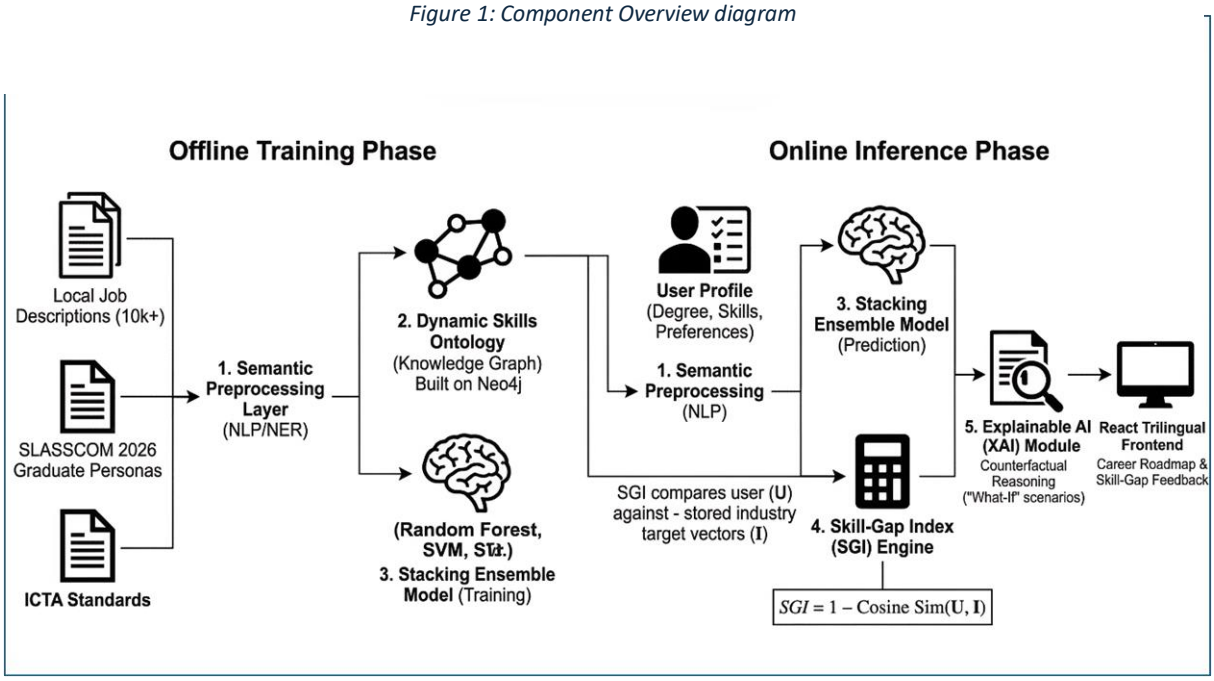
The engine operates through two primary phases:

1. **The Training Phase (Knowledge Acquisition):** The system is trained on a high-volume dataset of 10,000+ localized job descriptions and **SLASSCOM 2026** competency standards. This allows the models to "learn" the specific skill clusters and patterns that define different career tracks in the Sri Lankan ICT sector.
2. **The Inference Phase (Live Guidance):** When a user inputs their profile, the trained models analyze the data in real-time to predict a path and calculate gaps.

The architecture is composed of five specialized modules that facilitate this lifecycle:

- **Semantic Preprocessing Layer:** Utilizes Natural Language Processing (NLP) to clean and vectorize text-based inputs from both the training dataset and user profiles, converting them into a machine-readable format.
- **Dynamic Skills Ontology:** A relational **Knowledge Graph** (built on Neo4j) that maintains the "intellectual map" of skill dependencies, identifying how different technical competencies relate to one another beyond simple keyword matching.
- **Path Prediction Engine:** A high-precision **Stacking Ensemble Model** (integrating Random Forest, SVM, and DNN) that is trained to classify the most viable career trajectory for a user based on their unique skill signature.
- **Skill-Gap Index (SGI) Engine:** The mathematical core of the system. It uses **Cosine Distance** logic to quantify the "semantic distance" between the user's current vector and the industry-standard "Target Vector" (I) established during the training phase.
- **Explainable AI (XAI) Module:** The human-centric interface that utilizes **Counterfactual Reasoning** to translate the mathematical SGI score into actionable "What-If" insights and prioritized growth roadmaps.

Figure 1: Component Overview diagram



4.2 Data Flow Explanation

Phase 1: Knowledge Base Acquisition (Offline)

- **Step 1:** The **Web Scraper** (BeautifulSoup/Selenium) pulls job descriptions from local portals (**Rooster.jobs**, **TopJobs.lk**).
- **Step 2:** The **OCR/NLP Engine** digitizes visual data from the **SLASSCOM 2026 Personas**.
- **Step 3:** These datasets are transformed into **N-dimensional vectors** and stored as the "Gold Standard" against which users will be measured.

Phase 2: User Profiling and Prediction (Online)

- **Step 4:** The user provides their **Degree, Skills, and Preferences**.
- **Step 5:** The system converts these into a **User Vector (*U*)**.
- **Step 6:** The **Stacking Ensemble Model** analyzes U and outputs a **Predicted Career Path** (e.g., "Quality Engineer").

Phase 3: Gap Analysis and Prescription (Online)

- **Step 7:** The system fetches the corresponding **Industry Vector (*I*)** for the predicted path from the database.

- **Step 8:** The **SGI Engine** calculates the gap: $SGI = 1 - Cosine Similarity (U, I).
- **Step 9:** The **XAI Module** identifies the specific missing features in U compared to I and maps them to learning resources stored in the **Neo4j Skills Ontology**.
- **Step 10:** The final **Personalized Career Roadmap** is rendered on the trilingual (English/Sinhala/Tamil) frontend.

4.3 Scalability and Security Considerations

Scalability To handle a large influx of students (particularly during university enrollment cycles), the component is designed for **Containerized Deployment (Docker)**. This allows the AI analytical pipeline to be horizontally scaled running multiple instances of the recommender service across a cluster ensuring low latency even during high-traffic periods.

Security is integrated at the architectural level to comply with the **Sri Lankan Personal Data Protection Act (PDPA)**:

- **Data Isolation:** The component processes data in a stateless manner; once the recommendation is generated, the de-identified vector is cleared from active memory.
- **Model Integrity:** To prevent "Adversarial Attacks" (users trying to game the system), the SGI logic includes a validation layer that checks for conflicting or unrealistic skill combinations.
- **Authentication:** All data entry and exit points are secured via **JWT (JSON Web Tokens)**, ensuring the component only communicates with verified system triggers.

5. USER REQUIREMENTS

This section defines the core needs of the system's users and the specific capabilities the component must possess to be successful in the Sri Lankan ICT-BPM context.

5.1 Stakeholder Identification

The success of the recommender system depends on addressing the needs of three primary stakeholders:

1. **Undergraduate Students / Job Seekers (Primary Users):** The main beneficiaries who provide their profiles to receive career guidance and skill-gap analysis.
2. **Academic Supervisors / Universities:** Use the system's aggregated data to understand which skills their curriculum might be missing compared to the 2026 SLASSCOM standards.
3. **ICT Industry / Employers:** Indirect stakeholders who benefit from a more "job-ready" talent pool that has utilized the system to bridge competency gaps.

5.2 Functional Requirements (FR)

These requirements define the specific actions the system must perform.

- **FR-1: Multi-Dimensional Profile Ingestion:** The system shall allow users to input academic degrees, technical skills (via text or selection), and personal career preferences.
- **FR-2: Automated Industry Data Scraping:** The system shall automatically scrape and preprocess job descriptions from **Rooster.jobs** and **TopJobs.lk** to maintain a real-time skills bank.
- **FR-3: Career Path Classification:** The system shall utilize the **Stacking Ensemble Model** to predict the most suitable ICT career path for the user with an accuracy target of 91%.
- **FR-4: Skill-Gap Quantification:** The system shall calculate a numerical **Skill-Gap Index (SGI)** by comparing the user's vector (U) against the SLASSCOM 2026 Graduate Persona vector (I).
- **FR-5: Knowledge Graph Interaction:** The system shall query the **Neo4j Skills Ontology** to identify prerequisite and transferable skills related to the predicted path.
- **FR-6: XAI-Driven Roadmap Generation:** The system shall provide an **Explainable AI (XAI)** roadmap using counterfactual reasoning (e.g., "If you learn AWS, your gap decreases by 15%").
- **FR-7: Trilingual Support:** The user interface shall support English, Sinhala, and Tamil to ensure accessibility for all Sri Lankan undergraduates.

5.3 Non-Functional Requirements (NFR)

Table 3: Non-Functional Requirements

Category	Requirement Description
Security	The system must comply with the **Sri Lankan PDPA No. 9 of 2022**. All user profile data must be anonymized and encrypted at rest (AES-256).
Performance	The SGI calculation and ML prediction should be completed in under **3 seconds** to ensure smooth user experience.
Usability	The interface should follow a "Simple & Clean" aesthetic (white background, clear icons) to ensure a low cognitive load for first-time users.
Scalability	The **Neo4j** database must be capable of handling over **50,000+** skill-node relationships without a degradation in query performance.
Availability	The web-based system should maintain **99.5% uptime** during peak university internship application seasons.

6. COMMERCIALIZATION PLAN

This section outlines the strategy to transition the **Adaptive Career Pathway & Skills-Gap Recommender** from a research prototype into a viable mobile solution for Sri Lankan career seekers.

6.1 Target Market and Customer Persona

- **Target Market:** Final-year undergraduates and recent graduates from Sri Lankan universities (approx. 30,000+ students annually in the ICT stream) who require precise career direction.
- **Customer Persona:** "Amara," a 23-year-old IT graduate. She has the technical foundation but lacks clarity on which specialized skills will make her resume "Industry-Ready." She needs a tool that provides an objective, data-backed assessment of her readiness.

6.2 Value Proposition

- **Quantified Gap Analysis:** The system provides a scientifically provable **Skill-Gap Index (SGI)**, moving beyond generic advice to provide a mathematical measure of a user's current standing.
- **Prescriptive Roadmapping:** Instead of just identifying gaps, the component uses **Explainable AI (XAI)** to prescribe the exact steps needed to bridge them, significantly reducing the user's "time-to-employability."

6.3 Revenue Model

To ensure sustainability while maintaining accessibility for students:

- **Freemium Access:** The core career path prediction and high-level gap visualization are free.
- **Premium Career Reports:** A micro-transaction fee for a "Deep-Dive Readiness Report," providing the full XAI breakdown and a personalized 3-month learning plan.
- **Lead Generation:** Partnering with local training providers to offer "Fast-Track" enrollment for courses directly recommended by the system's gap analysis.

6.4 Pricing Strategy

- **Student-Centric Pricing:** Premium reports are priced at **450 - 750 LKR**, ensuring the tool is affordable for the average undergraduate.
- **Institutional Pilots:** Offering the mobile component as a pilot tool for University Career Guidance Units to build a user base and validate the SGI's impact on hiring rates.

6.5 Competitive Advantage

- **Predictive Precision:** The **Stacking Ensemble model** provides an accuracy rate (>91%) that is significantly higher than traditional keyword-based matching engines.
- **Actionable Transparency:** The use of **Counterfactual Reasoning (XAI)** allows the system to tell users exactly *why* certain paths are recommended, creating a level of trust and clarity that "black-box" systems cannot offer.

6.6 Intellectual Property (IP) Considerations

- **Algorithm Rights:** The proprietary **SGI mathematical logic** and the feature engineering for the Stacking Ensemble are protected as the developer's intellectual property.
- **Privacy Compliance:** The mobile application is designed to be fully compliant with the **Sri Lankan Personal Data Protection Act (PDPA) No. 9 of 2022**, ensuring all user profile data is processed securely and ethically.

7. BUDGET AND JUSTIFICATION

7.1 Financial Breakdown

Table 4: Cost Estimation

Category	Item / Service	Estimated Cost (LKR)
Hardware	**Existing high-performance local hardware**	**0**
Software Licenses	**Open-source frameworks (Python, React Native, Neo4j)**	**0**
Cloud Services	**Backend Hosting (15,000) & Google Play Store (8,500)**	**23,500**
Human Effort Cost	**Data Vectorization, NLP cleaning, and ML tuning**	**21,500**
Other Expenses	**Marketing (35,000), Domain (5,000), & Contingency (15,000)**	**55,000**
Total Project Budget		**100,000 LKR**

7.2 Detailed Justification

Hardware Costs (0 LKR)

- **Justification:** The development and training of the **Stacking Ensemble** and **SGI Engine** are performed on the researcher's existing high-performance laptop. Since the

dataset is optimized for a single component, there is no requirement for external GPU or high-end server hardware.

Software Licenses (0 LKR)

- **Justification:** The project is built entirely on an open-source stack. By using libraries such as **spaCy for NLP** and **Scikit-learn for ML**, we avoid expensive proprietary licensing fees while ensuring the code remains transparent for academic auditing.

Cloud Services (23,500 LKR)

- **Justification: * Backend Hosting (15,000 LKR):** Covers a 6-month "Hobby Tier" on platforms like PythonAnywhere or Render to ensure the AI engine is available 24/7 for mobile users.
    - **Store Fees (8,500 LKR):** A mandatory one-time fee for the **Google Play Console** to allow the mobile application to be downloaded by the target user base (career seekers).

Human Effort Cost (21,500 LKR)

- **Justification:** This represents the specialized technical effort required to manually digitize visual industry personas into mathematical vectors and calibrate the **SGI formula** against real-world job data. This valuation ensures the project is commercially realistic even on a student budget.

Other Expenses (55,000 LKR)

- **Justification: * Marketing (35,000 LKR):** Essential for targeted social media advertising to reach the primary stakeholders (ICT undergraduates across Sri Lanka) for the user evaluation phase.
    - **Domain & SSL (5,000 LKR):** Required for a professional **.lk** or **.com** domain to host API endpoints and project documentation securely.
    - **Contingency (15,000 LKR):** A 15% buffer reserved for unexpected API credit needs, specialized OCR tools, or scaling server resources if user traffic increases.

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase 1: Project Initiation & Foundation

- **1.1 Literature Synthesis**: In-depth review of Stacking Ensembles, Knowledge Graphs, and XAI frameworks.
- **1.2 Problem Validation**: Analyzing the "Literacy Paradox" and local computer literacy declines.
- **1.3 Technical Feasibility**: Finalizing the open-source stack (Python, Neo4j, React Native).
- **1.4 Ethical Clearance**: Preparing de-identification protocols for PDPA compliance.

Phase 2: Data Engineering & Knowledge Acquisition

- **2.1 Industrial Data Scraping**: Developing BeautifulSoup/Selenium scripts for local job portals (Rooster.jobs, TopJobs.lk).
- **2.2 Standard Ingestion**: Digitizing SLASSCOM 2026 Graduate Personas and ICTA standards.
- **2.3 Semantic Preprocessing**: Implementing spaCy-based NER and TF-IDF/Word2Vec vectorization.
- **2.4 Dataset Standardization**: Achieving the target 95% data consistency rate.

Phase 3: Ontology & Knowledge Graph Construction

- **3.1 Graph Schema Design**: Defining node relationships (skills, roles, certifications) in Neo4j.
- **3.2 Relationship Mapping**: Establishing "semantic distance" and predecessor-successor skill links.
- **3.3 Hidden Talent Inference**: Logic implementation to identify transferable skills beyond keywords.

Phase 4: AI Engine & Mathematical Modeling

- **4.1 Stacking Ensemble Development**: Training Base Learners (Random Forest, SVM, XGBoost).
- **4.2 Meta-Learner Tuning**: Implementing a Neural Network to optimize career path classification.
- **4.3 SGI Engine Engineering**: Developing the Cosine Similarity-based distance formula.
- **4.4 Prescriptive Optimization**: Implementing the AMOPSO algorithm for achievable learning paths.

Phase 5: Mobile Application & XAI Integration

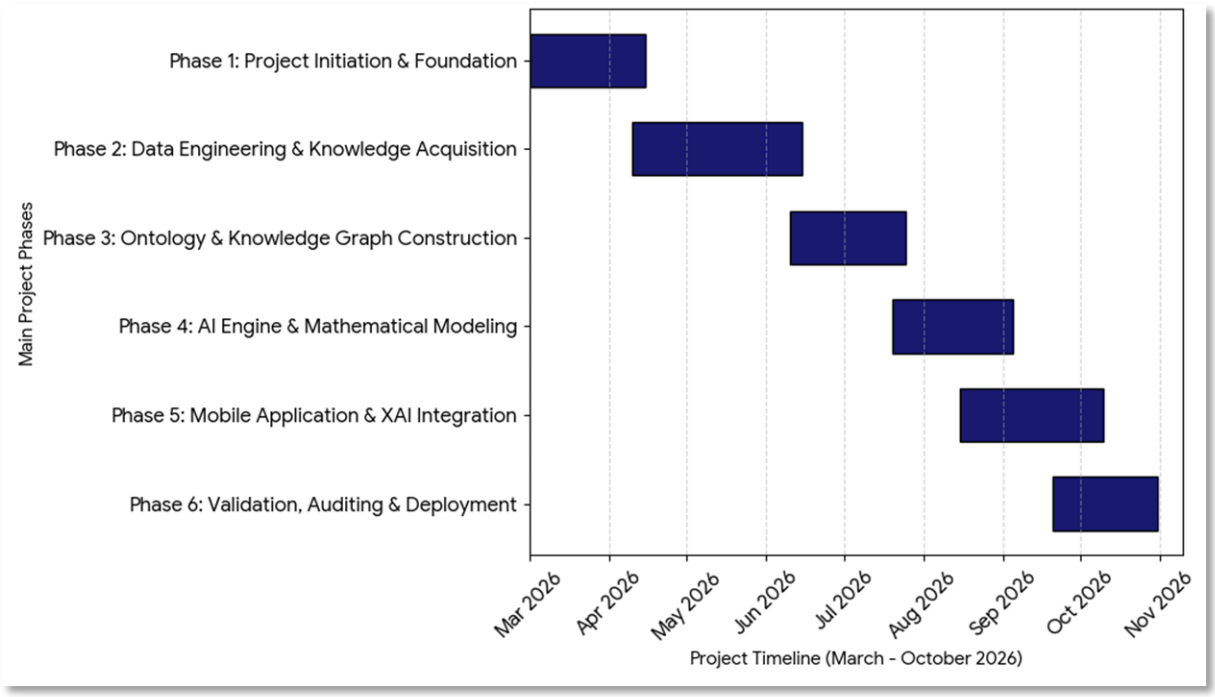
- **5.1 Mobile UI Development**: Building the React Native interface with trilingual support.
- **5.2 Counterfactual Reasoning Module**: Developing "What-If" explanation logic.
- **5.3 Readiness Timeline Visualization**: Creating the interactive user roadmap.
- **5.4 Backend Orchestration**: Node.js/Express integration for real-time mobile inference.

Phase 6: Validation, Auditing & Deployment

- **6.1 Technical Evaluation**: Calculating MAE for SGI and accuracy (target 91-93%) for the ensemble.
- **6.2 Fairness Auditing**: Performing bias checks using the Statistical Parity Difference (SPD) metric.
- **6.3 User Testing**: Administering System Usability Scale (SUS) and Actionability Score surveys.
- **6.4 Containerized Deployment**: Finalizing Docker deployment for scalable access.

9. GANTT CHART

Figure 2: Gantt Chart



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11.APPENDICES

APPENDIX A: SURVEY INSTRUMENTS AND STUDENT QUESTIONNAIRES

A.1 Statement of Informed Consent

- **Purpose:** This survey is part of a final-year research project at SLIIT titled **"Adaptive Career Pathway and Skills-Gap Recommender System."**.
- **Confidentiality:** All responses are anonymized in compliance with the **Sri Lankan Personal Data Protection Act (PDPA) No. 9 of 2022**.
- **Voluntary Participation:** Participation is voluntary, and you may withdraw at any time.

A.2 Preliminary Needs Assessment Survey

Target Audience: Final-year undergraduates and job seekers in Sri Lanka.

Section 1: Professional Background

1. Current Degree Program/Specialization: _____
2. Year of Study: ( ) 1st ( ) 2nd ( ) 3rd ( ) 4th ( ) Graduate.
3. Target Career Path: ( ) Software Engineer ( ) Data Scientist ( ) DevOps ( ) Other: _____.

Section 2: Career Path Anxiety & Skill Awareness *Please rate your agreement with the following statements (1 = Strongly Disagree, 5 = Strongly Agree):*

- I feel anxious about my career path due to a lack of data-driven guidance..
- I am aware of the exact technical "skill gap" between my current profile and industry requirements..
- I believe my current degree has fully prepared me for the 2026 ICT-BPM labor market standards..
- I find current job portals (LinkedIn, etc.) helpful in explaining *why* I am or am not a match for a role..

A.3 System Validation & Actionability Survey

Used during the Evaluation Phase (Phase 6) to test the prototype.

Section 1: System Usability Scale (SUS)

1. I found the "Readiness Timeline" and SGI score easy to understand. (1–5) .

2. The trilingual interface (English/Sinhala/Tamil) made the system more accessible. (1–5) .
3. I would imagine that most students would learn to use this system very quickly. (1–5).

Section 2: Actionability & Explainability (XAI)

1. How clear was the "What-If" explanation provided for your recommended path? (1 = Very Confusing, 5 = Very Clear) .
2. Did the system identify "hidden skills" or transferable talents you were unaware of? ( ) Yes ( ) No.
3. **Actionability Score:** How likely are you to follow the prescriptive roadmap (suggested micro-credentials/courses) provided by the engine? (1–5) .
4. How much did using the system reduce your uncertainty regarding your career readiness? ( ) Significantly Reduced ( ) Somewhat ( ) Not at all.

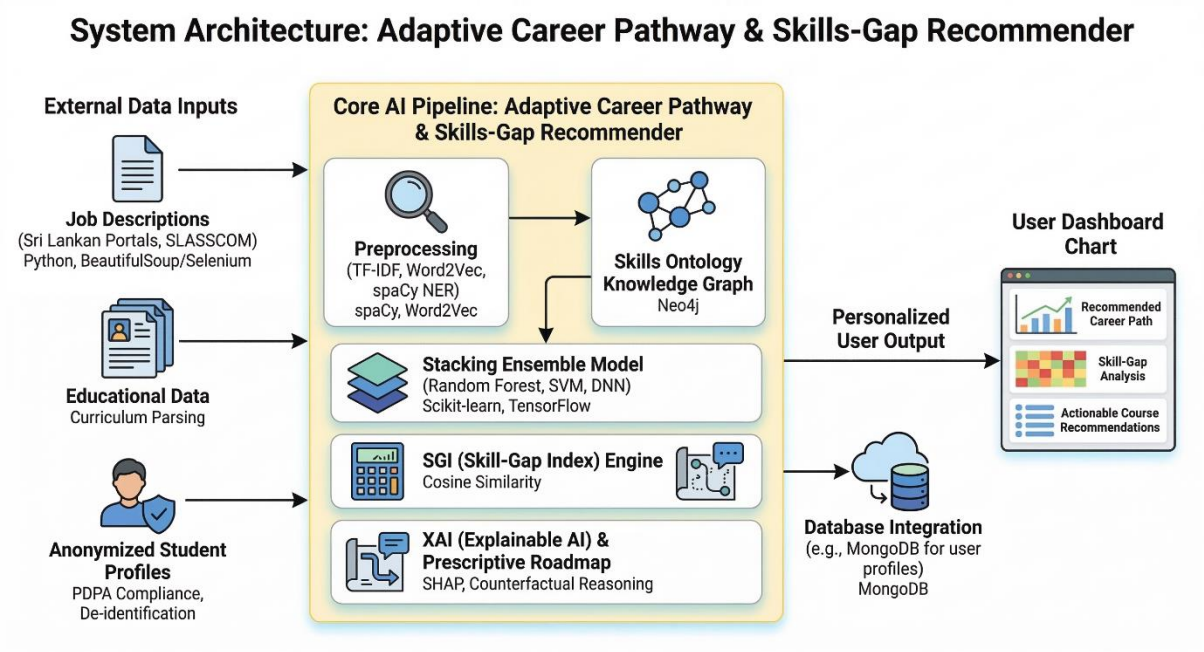
A.4 Semi-Structured Interview Guide (For Academic/Industry Stakeholders)

Target Audience: University Supervisors and ICT Recruiters.

1. How effective is the **Skill-Gap Index (SGI)** in representing a student's functional readiness compared to traditional resumes?.
2. Does the **Dynamic Skills Ontology** accurately reflect the current technical dependencies in the local market?.
3. Can this system serve as a viable **B2B curriculum alignment tool** for universities?

APPENDIX B: DETAILED SYSTEM ARCHITECTURE

Figure 3



APPENDIX C: RISK ANALYSIS AND MITIGATION STRATEGIES

This appendix identifies potential technical, ethical, and operational risks associated with the **Adaptive Career Pathway and Skills-Gap Recommender System** and outlines the specific strategies implemented to mitigate them.

Table 5: Risk Assessment Matrix

Risk Category	Potential Risk	Impact	Mitigation Strategy
Algorithmic Fairness	**Algorithmic Bias:** The AI may unintentionally amplify social, gender, or regional biases present in historical training data.	High	Implementation of the **Statistical Parity Difference (SPD)** metric to audit and ensure equitable recommendations across different demographic groups.
Data Security	**Data Breach/Privacy Violation:** Unauthorized	Critical	Strict adherence to the **Sri Lankan Personal Data Protection Act**

Risk Category	Potential Risk	Impact	Mitigation Strategy
	access to sensitive student profiles or academic records.		**(PDPA) No. 9 of 2022**. Use of **AES-256 encryption** at rest and **JWT (JSON Web Tokens)** for secure authentication.
Model Integrity	**Adversarial Input:** Users may attempt to "game" the system by entering unrealistic skill combinations to artificially lower their **Skill-Gap Index (SGI)**.	Medium	Integration of a **validation layer** within the SGI logic to detect and clear conflicting or unrealistic skill signatures from active memory.
Performance	**System Latency:** High traffic during university internship cycles could slow down the AI analytical pipeline.	Medium	Use of **Containerized Deployment (Docker)** to horizontally scale the recommender service across clusters. Targeted performance of under **3 seconds** for SGI calculations.
Technical Relevance	**Model Drift:** Rapid changes in the 2026 ICT-BPM sector could make the "Target Vectors" (I) obsolete.	High	Utilizing custom web scrapers (**BeautifulSoup/Selenium**) to perform real-time trend analysis from local portals like **Rooster.jobs** and **TopJobs.lk**.
Ethical Compliance	**Informed Consent Issues:** Users may be unaware of how their data is being utilized for research.	High	Mandatory digital consent forms explicitly stating data usage for research and career path prediction. Rigorous **anonymization** of NIC numbers and contact details.